

## ML / AI Tier

The ML tier (Tier 3) provides **advisory** anomaly detection, distribution drift monitoring, and a provenance-tracked model registry. It is implemented in `src/ml/` and surfaced under `/api/v1/ml`.

### The advisory-only guarantee

The ML/AI tier can inform operators. It can **never** command a mission-state change or alter Tier-1/Tier-2 data.

This is enforced structurally, not just by convention:

- `MLAnalysisWorker` holds **no reference** to the mission state machine.
- No file under `src/ml/` imports or references `MissionStateMachine` / `mission_machine`. This is asserted by `tests/test_ml.py::test_ml_source_does_not_reference_mis`
- Every ML API response carries an advisory note and cannot mutate mission state.

### Detectors

| Detector                      | Dependency        | Availability            | Method                                                                                                                                                         |
|-------------------------------|-------------------|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>robust_zscore</code>    | none (pure NumPy) | <b>Always available</b> | Median/MAD robust<br>z-score: $ x - \text{median}  / (1.4826 \cdot \text{MAD})$ ,<br>cutoff 3.5.<br>Zero-MAD features are guarded, not turned into infinities. |
| <code>isolation_forest</code> | scikit-learn      | If sklearn imports      | Isolation Forest,<br><code>contamination='auto'</code> ,<br>200<br>estimators.                                                                                 |
| <code>gmm</code>              | scikit-learn      | If sklearn imports      | Gaussian Mixture<br>negative log-likelihood,<br>5th-percentile<br>cutoff.                                                                                      |

| Detector                   | Dependency | Availability                              | Method                                                                                                                                                                                  |
|----------------------------|------------|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>lstm_forecast</code> | PyTorch    | If torch imports                          | One-step LSTM forecast; residual z-scored, cutoff 3.0.                                                                                                                                  |
| <code>ensemble</code>      | —          | Uses whichever of the above are available | Rank-normalises each available detector's scores, averages them, and flags a sample when a majority ( 50%) of contributors vote anomalous. Records exactly which detectors contributed. |

GET `/api/v1/ml/detectors` returns a **live** availability probe: for each detector it reports **available**, the **method**, and — when unavailable — the exact **reason** (e.g. the import error). Nothing is claimed to work that cannot.

## Feature extraction — no imputation

`extract_feature_matrix` builds the numeric matrix from a window of telemetry frames. A frame contributes a row **only if every requested feature is present and valid** in that frame. Incomplete frames are dropped and counted (`dropped_incomplete`); they are never zero-filled or imputed. The default feature set is:

```
bme688_temperature, bme688_pressure, bme688_humidity,
gnss_altitude, radiation_cpm, battery_voltage
```

## Drift monitoring

`population_stability_index(expected, actual)` computes the PSI between a reference distribution and a current window:

| PSI               | Classification                                                        |
|-------------------|-----------------------------------------------------------------------|
| < 0.10            | STABLE                                                                |
| < 0.25            | MODERATE                                                              |
| 0.25              | SIGNIFICANT                                                           |
| insufficient data | <b>UNAVAILABLE</b> (returns <code>None</code> , never a fabricated 0) |

The `DriftMonitor` stores a reference matrix and evaluates per-feature PSI on demand. Insufficient data yields an explicit unavailable result.

## Training

`POST /api/v1/ml/train` (SCIENTIST role, rate-limited 2/hour) runs `train_detector` over stored telemetry:

1. Reconstruct frames from persisted measurements.
2. Extract the feature matrix (dropping incomplete frames honestly).
3. Refuse to train if fewer than **20** valid samples — returns 422 with the exact count and how many frames were dropped. **No fake model is created.**
4. Fit and score the detector.
5. Persist the artefact via `ModelStore` and record its provenance.

## Model provenance & integrity

Every trained model records:

- `training_data_hash` — a hash of the exact feature matrix it was trained on.
- `sha256_hash` — the SHA-256 of the serialised artefact on disk.

`ModelStore.load(path, expected_sha256)` **recomputes** the file's SHA-256 and refuses to load if it is missing, if no expected hash is supplied, or if the hash does not match. A tampered model file therefore fails closed with a `SafeLoadError` rather than executing an untrusted pickle. This is verified by `tests/test_ml.py::test_model_store_detects_tampering`.

## Live advisory summary

The `MLAnalysisWorker` maintains a rolling window and periodically scores it with the ensemble. `GET /api/v1/ml/anomalies` returns the latest summary with an honest state:

| State                     | Meaning                                       |
|---------------------------|-----------------------------------------------|
| IDLE                      | Worker running, no analysis yet.              |
| INSUFFICIENT_DATA         | Fewer than the minimum samples in the window. |
| UNAVAILABLE               | No detector could run.                        |
| AVAILABLE                 | A real anomaly summary is present.            |
| NOT_CONFIGURED (HTTP 503) | The ML worker is not running at all.          |

At no point does the tier invent a “clean” result to appear healthy.